ChurnPulse · Quick Guide

Predict which customers will leave, see why, and know what to offer — live.

What it does

ChurnPulse scores every customer's chance of leaving, using five kinds of data — **where they live, who they are, the tech they use, how they behave and how they feel**. It explains every score and suggests the best action to keep them.

85.5%
accurate from a customer's profile alone (any new member)

96.5%
accurate when a satisfaction survey is known*

1 Start it

Double-click **ChurnPulse** on your desktop. Or in a terminal:

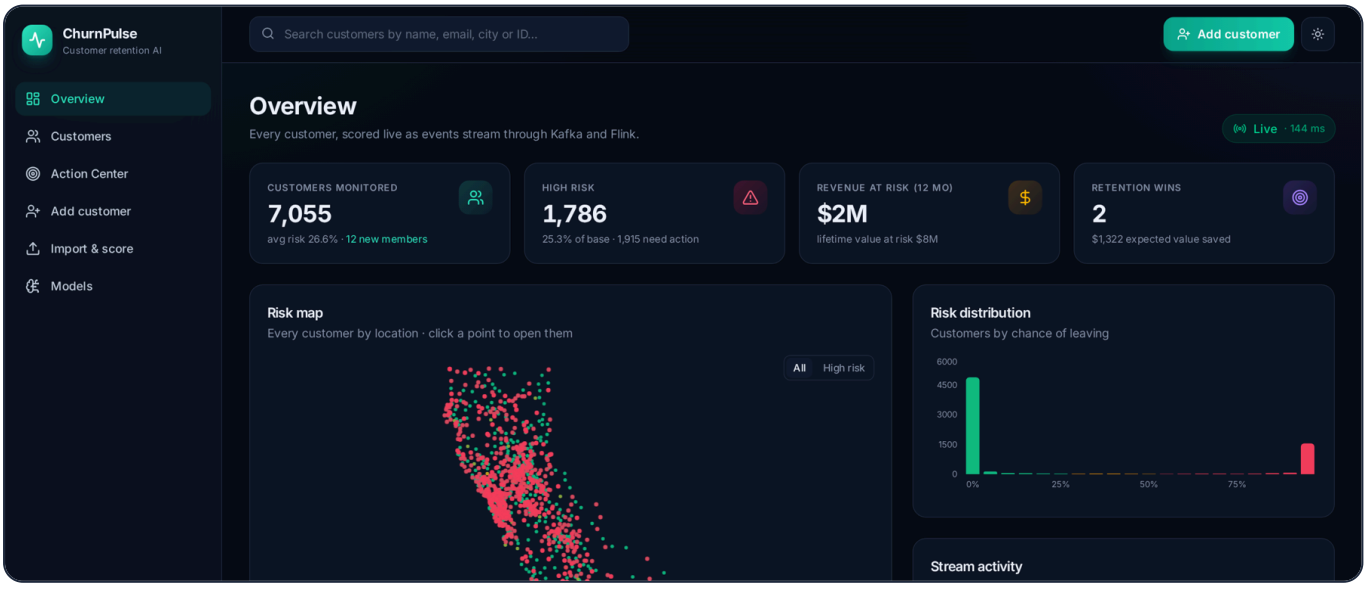
```
cd ~/Desktop/ChurnPulse
make up
```

⏱ About 30 seconds (2 minutes the first time). The dashboard opens by itself at **localhost:3000**.

2 Stop it

Double-click **Stop ChurnPulse**, or run `make down`. Your data is kept for next time.

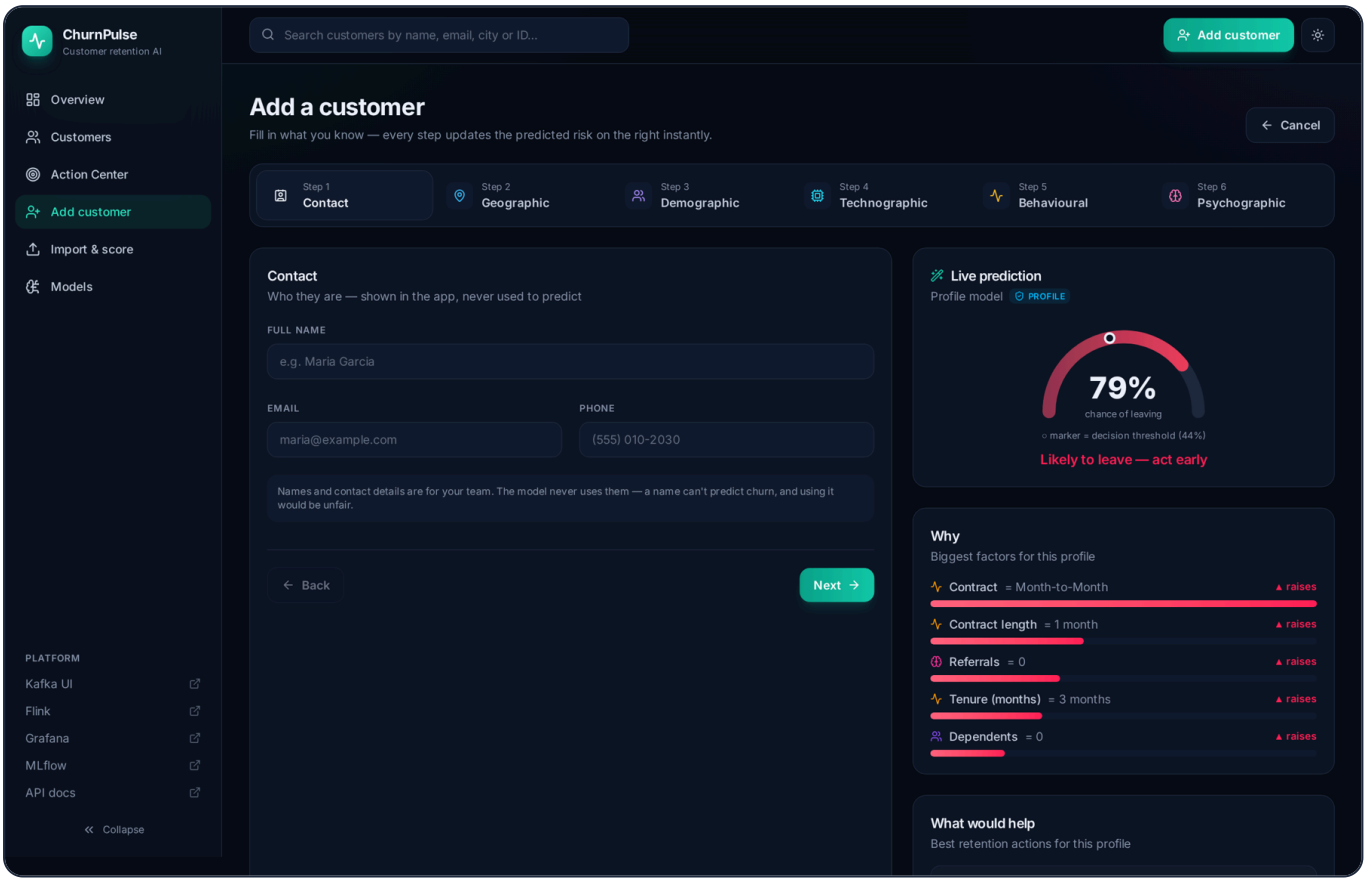
3 The Overview



Key numbers, a risk map of every customer (click a dot to open them), and live predictions as customers change. Switch light/dark with the ☀/☾ button, top right.

➕ Add a new customer and predict them

1. Click **Add customer** (top right).
2. Fill in the 6 steps: **Contact → Geographic → Demographic → Technographic → Behavioural → Psychographic**. Type a city name — location details fill in automatically.
3. Watch the **live prediction** on the right change as you type.
4. Press **Add customer**. They're saved and tracked live from now on.

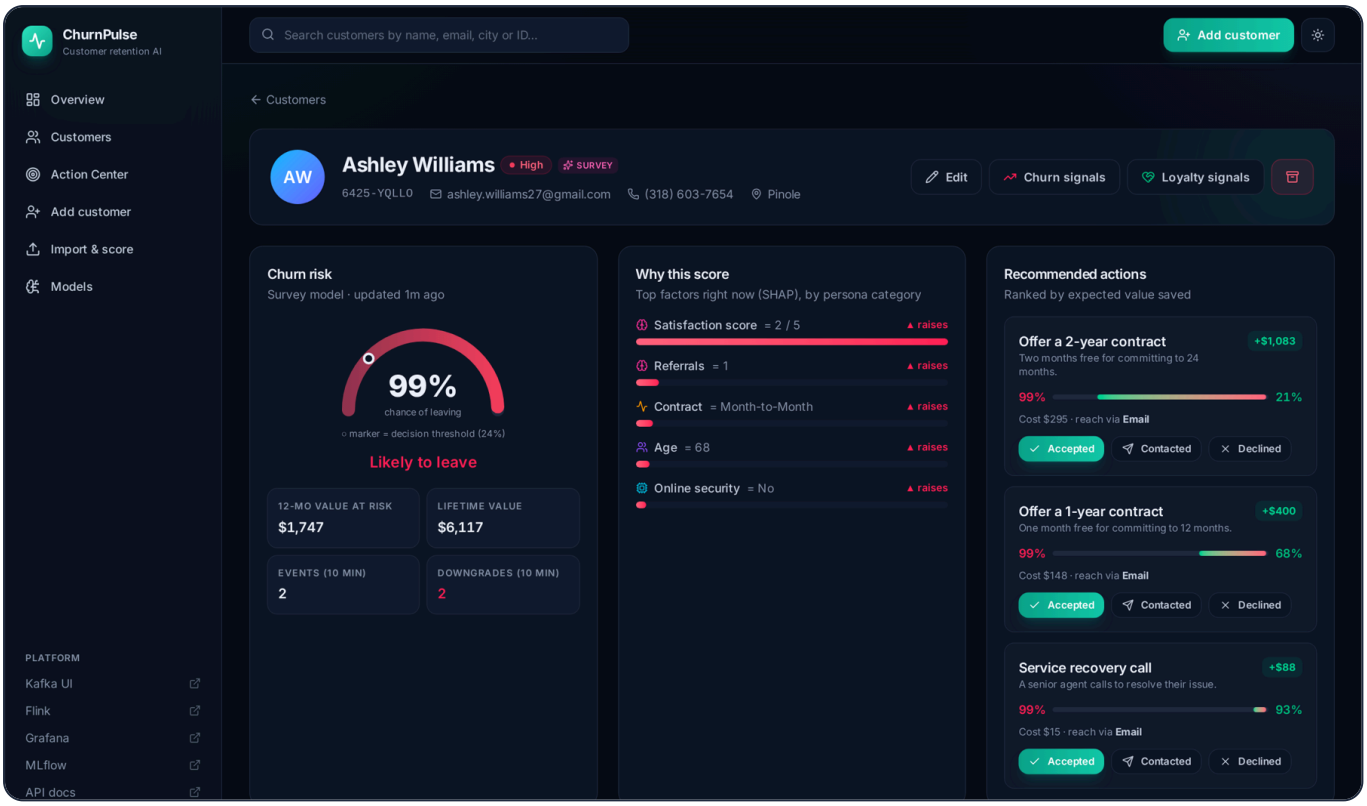


💡 **Satisfaction survey known?** Pick a score 1–5 in the last step and the more accurate survey model is used. Leave it as “Not surveyed” and the profile model is used.

Many customers at once

Import & score in the menu → download the CSV template → fill it in (Excel works) → drop it on the page. Choose **Import as customers** to save them, or **Score only & download** to get a scored spreadsheet back.

👤 A customer's page



- **Churn risk** — the chance they leave, and which model scored them (PROFILE or SURVEY).
- **Why this score** — the main reasons, coloured by data type. Red raises risk, green lowers it.
- **Recommended actions** — the best offers with the money each is expected to save. Press **Accepted** when the customer takes it and watch their risk drop live.
- The tabs below show all their data by category, plus a **Timeline** of how their risk changed.
- **Churn signals** / **Loyalty signals** send demo events so you can see the live system react.

🎯 Action Center

The list of who to contact today, ranked by how much money it could save — with the best offer and how to reach each person (email, phone, SMS). Log every call as **Contacted**, **Accepted** or **Declined**.

If something doesn't work

PROBLEM	FIX
Page doesn't load	Wait 1 minute after starting, then refresh.
Overview says “Reconnecting…”	Run `make status`; if something stopped, run `make up`.
A new customer page shows loading	It is being scored live — it appears within a few seconds.
CSV import reports failed rows	The result lists each row and the problem (e.g. invalid email) — fix and re-import those rows.
PC feels slow	Stop ChurnPulse when not in use (it uses about 4.5 GB RAM).
Fresh start wanted	`make reset` then `make up` (deletes added customers and history).

Useful commands

COMMAND	WHAT IT DOES
`make demo`	Runs a churn example and prints before / after
`make test` · `make test-e2e`	Runs the 33 unit tests · the 6 live end-to-end tests
`make train`	Retrains both models (~6 minutes)
`.venv/bin/jupyter lab notebooks`	Opens the 4 notebooks (start with `01_eda`)

Good to know

* **About the 96.5%.** In the IBM dataset, satisfaction mirrors churn almost perfectly (everyone who scored 1–2 left; nobody who scored 4–5 did). Real surveys are weaker, so expect less on real customers. The profile model's 85.5% is the dependable figure.

Live customer changes on the dashboard are **simulated**. Names and contact details are generated and are never used to predict. Recommended-action estimates are model-based — logging real outcomes shows what truly works.